COMPRESSION DIAGNOSTICS Measuring What the AI Burns, Invents, and Distorts

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Abstract

Compression Diagnostics defines a quantitative practice for evaluating entity representation after AI compression. It adapts the Three Compressions and the Encyclotron into five outputs: beige threshold, content gain or hallucination, content loss or erasure, semantic coherence or fragmentation, and an overall R1/R2/R3 regime classification. The framework is aimed at organizations and public entities whose descriptions may be technically accurate yet generic, fragmented, or stripped of differentiating intellectual capital.

A Basecamp audit is used as calibration, including a reported beige score of 0.71 and qualitative findings of high content loss, low invention, and disconnection among product, methodology, and founder. These values are outputs of the document's own audit method, not independently reproduced benchmarks. The practice is positioned as the measurement arm of a larger workflow: Retrieval Forensics investigates causes, Retrieval Architecture builds corrective structures, Entity Integrity handles disambiguation, and metadata packets deploy the resulting entity definition.

Body

COMPRESSION DIAGNOSTICS

Measuring What the AI Burns, Invents, and Distorts

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ABSTRACT

Compression Diagnostics is the quantitative measurement of what survives AI compression. Using the Three Compressions framework (R1: Lossy/Commoditization, R2: Predatory/Capital Erasure, R3: Witness/Semantic Sovereignty) and the Encyclotron diagnostic instrument, Compression Diagnostics produces numerical scores for entity flattening, content loss, hallucination, and semantic fragmentation in the AI retrieval layer. This document defines the measurement framework, specifies the metrics, and presents calibration data.

THE MEASUREMENT GAP

Every discipline needs measurement. Medicine has bloodwork. Engineering has stress tests. Finance has audits. The AI retrieval layer — the infrastructure that now determines how

entities are discovered, described, and attributed — has no established measurement science.

SEO measures rankings and traffic. GEO measures citation frequency. Neither measures the thing that matters: **what happens to your entity's meaning when the AI compresses it.**

When the AI summarizes your organization into 4–5 citations and ~169 words, it makes decisions about what to preserve and what to burn. Those decisions determine whether your differentiation survives, whether your IP is attributed, and whether a prospect has a reason to choose you over a competitor. No existing tool measures these decisions.

Compression Diagnostics measures them.

THE METRICS

Compression Diagnostics produces five quantitative metrics per entity: ### β — Beige Threshold (0.0 – 1.0)

The proportion of the AI's description that could apply to any competitor in the same category. Measures entity-level genericness.

Score	Interpretation
0.0 – 0.3	**Distinctive.** Description captures what makes you different.
0.3 – 0.6	**Partial differentiation.** Some specifics, some generic language.
0.6 – 0.8	**Commodity zone.** Most of description fits any competitor.
0.8 – 1.0	**Placeholder noun.** Entity has ceased to exist as a distinct representation.

Calibration: Basecamp (37signals) scored β = 0.71 — commodity zone. 71% of the AI's description could apply to Monday.com, Asana, or ClickUp. ### Δ_G^+ — Content Gain (Hallucination Index)

What the AI invented that does not exist. Measured in distinct false claims per diagnostic level. Low Δ_G^+ means the AI is not hallucinating about you. This is typically good — unless the hallucinations are *favorable* extensions of your frameworks (see: Conceptual Infrastructure Ownership, EA-CORP-04). ### Δ_G^- — Content Loss (Erasure Index)

What the AI dropped that matters. Measured as the number of differentiation-critical attributes absent from the AI's description. High Δ_G^- means your competitive advantage is invisible.

Calibration: Basecamp's Δ_G^- was HIGH — six differentiation-critical attributes (calm company philosophy, intentional simplicity, Shape Up as competitive advantage, bootstrap trust signal, founder thought leadership, HEY email as vision evidence) were absent from all commercial queries. ### S_c — Semantic Coherence (Fragmentation Score)

Whether the AI treats your entity as one coherent thing or as disconnected fragments. Measured as the number of entity-level disconnections across diagnostic levels.

Calibration: Basecamp showed S_c = FRAGMENTED — the product, methodology, and founder were retrievable as three separate entities but never connected in commercial queries. ### R — Compression Regime (R1 / R2 / R3)

The classification of the compression behavior the entity is experiencing, per diagnostic level and overall:

Regime	Behavior	Revenue Impact
R1	Commoditization — flattened to consensus	Brand equity eroding
R2	Capital Erasure — value extracted without credit	IP being consumed
R3	Semantic Sovereignty — meaning survives intact	Market position defended

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THE INSTRUMENT: THE ENCYCLOTRON

The Encyclotron (DOI: 10.5281/zenodo.19474724) is the diagnostic instrument that produces Compression Diagnostics measurements. It runs 45 structured queries across five diagnostic levels (Entity Recognition, Competitive Position, Intellectual Property, Customer Decision, Founder Entity) and scores each for β, Δ_G^+, Δ_G^-, S_c, and R.

No other instrument in the GEO/AEO/SEO industry measures compression behavior. No other instrument has a DOI-anchored methodology published on CERN's Zenodo.

APPLICATIONS

Corporate audits: Baseline measurement before and after Retrieval Architecture interventions. **Competitive analysis:** Comparative β scores across entities in a category. **IP monitoring:** Tracking Δ_G^- over time to detect emerging attribution failure. **Due diligence:** Compression regime mapping for acquisitions (is the target's brand R1 or R3?). **Regulatory compliance:** Documented evidence of AI misrepresentation for legal proceedings.

RELATED DISCIPLINES

Compression Diagnostics is the measurement dimension of a unified methodology. Retrieval Architecture (EA-RA-01) provides the constructive interventions based on diagnostic findings. Retrieval Forensics (EA-RFO-01) investigates the causes of compression damage. Entity Integrity (EA-EI-01) addresses disambiguation specifically. Metadata Packets for AI Indexing (EA-META-01) specify the deployable artifact.

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